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COVER LETTER

Dear Hiring Manager,

In my search for a new challenge where I can leverage my knowledge in Data Engineering, I was pleased to find Barclays's opening for a Senior Data Engineer role and I believe my recent education and experience in data engineering closely match your requirements.

During the 8 years of experience in data engineering domain, I learned the principles of Data Engineering, advanced Python programming, AWS Cloud, SQL, Data Pipelines, Big Data ecosystem (Hive, Spark, Sqoop, MR, HDFS, Oozie), and much more. I understand that for this position knowledge of big data frameworks, Internals of Apache spark and job orchestrators is required which I have experience in all of these. I built a few Data Pipelines in the on-prem cloudera 25 node cluster with 8TB of memory and also few pipelines using the AWS cloud components such as S3, Lambda, SQS, Cloud Watch, Secrets Manager and EMR.

here are few things that I would like to highlight myself in terms of data delivery and accomplishments:

- Automated ETL processes, making it easier to wrangle data and reducing time as much as 35%.
- Ensured end to end process of data extraction until reporting and noted efficiency of close to 20%.
- Ingested data from 12 different data sources from the upstreams comprising of data logs from the vendors.
- sqoop is used to ingest the tagging data into HDFS in every 15 minutes.
- creation of sqoop job resulted in 15% reduce in latency in ingesting the data into the system.
- designed and built data pipeline to ingest tagging data of 150 million records from 9 plus sources using pyspark.
- Migrated slow processing ETL pipelines to spark which resulted in 72% faster execution of pipelines and 45% more realibility in data delivered.

Few Spark Optimization Techniques I follow:

- Partitioning & Bucketing
- Cache & Persist
- Coalesce & Repartition
- Spark Joins (SMB, Broadcast)
- ReduceByKey vs GroupByKey
- Distributed Storage
- In-Memory Processing
- Right File Formats (Choice of Avro, ORC, Parquet)
- Balanced Cluster Resource Utilization
- Avoiding Garbage Collection
- Avoiding Data Shuffle Across the cluster
- Sort vs Hash Aggregate
- Choosing between RDDs, Dataframes, Datasets
- Static vs Dynamic Resource Allocation
- Even Data Distribution, Handling Skewed Partiitons, Salting, Increasing Spark Degree of Parallelism.

Barclays appeals to me as the perfect company to further enhance my data engineering career and I'm sure that Barclays's data team with experienced data scientists and engineers enables me to hone my skills and allow me to contribute to the future success of the company.

I would welcome the opportunity to have an interview and discuss further and I look forward to hearing from you!

Sincerely,
Kiran Gunturu